

Representation Separability of Signal-to-Image Transformations for Sleep Stage Classification Using Pretrained Vision Transformers

Accurate sleep stage classification is essential for diagnosing sleep disorders, guiding treatment decisions, and advancing the understanding of brain function during sleep. Manual scoring of polysomnography (PSG) remains the clinical gold standard but is time-consuming and prone to inter-rater variability, motivating the need for reliable automated methods. While convolutional neural networks have demonstrated strong performance in this domain, Vision Transformers (ViTs) remain underexplored despite their ability to capture global dependencies. The primary objective of this study is to systematically evaluate the effectiveness of different signal-to-image (S2I) transformations for ViT-based sleep staging and to identify strategies that yield the most discriminative representations.

This study systematically evaluated 1,016 combinations of S2I methods, fusion strategies, and channel selections using the CAP Sleep Database, comprising 108 overnight PSG recordings annotated into standard 30-second sleep epochs. After preprocessing and artifact removal, four S2I transformations (spectrograms, GASF, GADF, and MTF) were applied, with fusion achieved through grid tiling or channel stacking. Embeddings extracted from a finetuned ViT-B/16 served as the basis for assessing stage separability using Fisher's Discriminant Ratio, Silhouette Coefficient, Davies–Bouldin Index, and Calinski–Harabasz Index. Composite scores derived from these metrics ranked strategies, and the top were validated through classification benchmarks with several base ViT models.

Results showed that GASF with grid tiling across all channels produced the most discriminative representations (composite score = 3.95), significantly outperforming other strategies (Mann Whitney U, $p < 0.001$, $r = 0.68$, where p denotes significance level and r denotes rank-biserial correlation effect size). This top-ranked transformation was further assessed using five Vision Transformer architectures (ViT-B/32, DeiT-B, Swin-B, BEiT-B, and PVT-Small) under five-fold stratified cross-validation. Performance was evaluated by overall accuracy and macro-averaged F1-score. ViT-B/32 achieved the highest results (83.1% accuracy, 79.4% macro-F1), with Swin-B and BEiT-B performing closely. These findings confirm that effective S2I strategies enhance ViT-based sleep staging and underscore the clinical potential of Vision Transformers.

Keywords: Sleep Stage Classification, Vision Transformers, Signal-to-Image Transformation, Deep Learning

Introduction

Sleep is a fundamental physiological process essential for memory, cognition, and health. Accurate characterization of sleep stages is therefore vital for diagnosing disorders and advancing the understanding of brain function during sleep (Iber et al., 2007). Manual scoring of polysomnography (PSG) remains the gold standard but is time-consuming and subjective, motivating research into automated classification methods.

Deep learning has significantly advanced this field. Convolutional neural networks (CNNs) have achieved strong results by capturing local spatiotemporal patterns in EEG signals (Phan et al., 2019). Recent work has shown that transforming EEG epochs into time–frequency images, such as spectrograms or wavelet representations, further enhances CNN-based classification (Rahman et al., 2022). Building on these successes, Vision Transformers (ViTs) have emerged as a powerful alternative to CNNs, leveraging self-attention to model global dependencies in data (Dosovitskiy et al., 2021). Recent studies, such as SleepDIFFormer, demonstrate the promise of ViT architectures for explainable and accurate PSG-based sleep staging (Li et al., 2025).

A critical challenge in adapting ViTs to EEG is the conversion of one-dimensional signals into two-dimensional image representations. Spectrograms capture rich frequency features but can disrupt temporal continuity, while Gramian Angular Fields (GASF, GADF) and Markov Transition Fields (MTF) preserve temporal correlations and nonlinear dynamics (Wang & Oates, 2015). Identifying the most effective signal-to-image (S2I) transformation is therefore essential.

This study systematically evaluates multiple S2I strategies for PSG-based sleep staging with Vision Transformers. It compares their ability to preserve stage separability, assesses discriminative quality through statistical evaluation, and investigates whether the most effective representations improve downstream classification performance.

Methodology

This study employed a structured experimental pipeline to evaluate different S2I transformation strategies for sleep stage classification using ViTs. The methodology consisted of three main phases: signal preprocessing, S2I transformation and fusion, and evaluation based on representation separability and classification performance.

Experiments were conducted using the publicly available CAP Sleep Database (Goldberger et al., 2000), which contains 108 overnight PSG recordings ranging from 3 to 9 hours in duration. The database includes recordings from both healthy participants and patients with sleep disorders such as insomnia, narcolepsy, and sleep-disordered breathing. Each recording provides multiple physiological channels, including electroencephalography (EEG), electrooculography (EOG), electromyography (EMG), electrocardiography (ECG), airflow, respiratory effort, and oxygen saturation. Sleep stages were annotated in 30-second epochs by clinical experts according to the Rechtschaffen and Kales criteria (Rechtschaffen & Kales, 1968).

Preprocessing was conducted to prepare the signals for subsequent transformations and classification. Filtering was applied on a modality-specific basis: EEG and EOG were band-pass filtered between 0.3–35 Hz, ECG between 0.3–70 Hz, and EMG between 10–70 Hz, consistent with prior PSG preprocessing standards (Iber et al., 2007). A 60 Hz notch filter was employed to suppress powerline interference. Artifact detection was performed using a combination of amplitude thresholds, flatline detection, and line-noise ratio measures. Identified artifact segments were excluded from further analysis to ensure data quality.

Each channel was segmented into 30-second epochs and transformed into two-dimensional representations using four methods: short-time Fourier transform (STFT)-based spectrograms, Gramian Angular Summation Field (GASF), Gramian Angular Difference Field (GADF), and Markov Transition Field (MTF). Spectrograms were computed with 2-second Hann windows and 50% overlap, retaining temporal–frequency resolution. GASF and GADF were generated by rescaling signals into $[-1, 1]$, mapping them into polar coordinates, and computing Gramian matrices (Wang & Oates, 2015). MTF representations were created by discretizing signals into 16 quantile bins, computing transition probabilities, and projecting them across time indices (Wang & Oates, 2015).

To incorporate multi-channel information, two fusion strategies were evaluated. In grid tiling, channel-specific images were arranged into spatial grids (e.g., 2×2 for four channels), with empty positions zero-padded. In channel stacking, channels were mapped to RGB dimensions; for higher-dimensional data, principal component analysis was used to reduce to three orthogonal components, following similar strategies in multimodal EEG studies (Zhang et al., 2019). All composite images were resized to 224×224 pixels to match the input resolution required by ViTs (Dosovitskiy et al., 2021).

To preliminarily assess the quality of S2I representations, a finetuned pretrained ViT-B/16 initialized with ImageNet-1K weights was employed as a feature extractor (Dosovitskiy et al., 2021). For each transformed 30-second epoch, the [CLS] token from the final transformer block was extracted and treated as the embedding representation. These embeddings served as the basis for evaluating how well different S2I transformations preserved sleep stage separability.

Quantitative analyses were used to evaluate discriminative quality using Fisher’s Discriminant Ratio (FDR) which was computed to measure the trade-off between inter-class variance and intra-class compactness (Liu & Yu, 2005). In addition, widely used cluster validity indices including the Silhouette Coefficient (Rousseeuw, 1987), Davies–Bouldin Index (Davies & Bouldin, 1979), and Calinski–Harabasz Index (Calinski & Harabasz, 1974) were calculated using ground truth sleep stage labels, providing complementary insights into cluster separation. Together, these metrics offered a robust assessment of linear and nonlinear class separability across transformations and fusion strategies.

Tested 1016 different combinations of above mentioned channels, S2I transformation and fusion strategies. For each of the 1,016 combinations, 1,000 bootstrap samples of embeddings were generated, and all four evaluation metrics were computed per sample. Mean values were then normalized via min–max scaling and combined into a composite score: normalized FDR + normalized Silhouette + normalized CHI – normalized DBI. In cases of ties, FDR was prioritized due to its direct focus on separability. Strategies were ranked by composite score, and the top five were selected for downstream classification experiments.

To determine whether the top-ranked S2I strategies outperformed other approaches, composite scores were compared using the Mann–Whitney U test, with effect sizes quantified via rank-biserial correlation (Fagerland, 2012). When overall significance was observed, pairwise comparisons were restricted to the top five ranked strategies, as these represented the most competitive candidates while ensuring computational feasibility and interpretability. The Wilcoxon signed-rank test was applied with Bonferroni correction for multiple comparisons, and Cohen’s d was reported as the effect size. The top-ranked transformation was then evaluated in a downstream classification benchmark using different Vision Transformer architectures, including ViT-B/32, DeiT-B, Swin-B, BEiT-B, and PVT-Small. Five-fold stratified cross-validation ensured balanced sleep stage representation, and performance was measured via overall accuracy and macro-averaged F1-score. This evaluation verified that strategies yielding high separability also translated into robust classification performance, supporting their practical utility in automated sleep stage analysis.

Results and Discussion

The results of the top 10 tested combinations of S2I transformations, fusion strategies, and channel selections are reported in Table 1. The table shows that GASF with grid tiling across all channels achieved the highest composite score (3.95), indicating superior separability of sleep stages. GADF with grid tiling also performed strongly (3.63), while spectrograms and MTF-based representations yielded comparatively lower scores, reflecting weaker discriminative capacity. Interestingly, strategies that incorporated all channels tended to perform better than those restricted to subsets (e.g., EEG + EOG + ECG), suggesting that broader multimodal information contributes positively to representation quality. Overall, the ranking confirms that Gramian Angular Field based methods, particularly GASF, are the most effective for generating ViT-compatible representations.

Table 1 Top 10 Combinations according to composite score

S2I Method	Fusion Strategy	Channels	Avg. FDR	Avg. Silhouette	Avg. DBI	Avg. CHI	Composite Score
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GASF	Grid Tiling	All	2.80	0.62	0.78	18573	3.95
GADF	Grid Tiling	All	2.45	0.54	0.95	16212	3.63
GASF	Channel Stacking	All	2.10	0.47	1.10	14055	3.21
MTF	Channel Stacking	All EEGs + EOG + ECG	1.85	0.31	1.45	11592	2.77
GASF	Channel Stacking	All	1.60	0.18	1.80	9809	2.23
Spectrograms	Grid Tiling	All	1.30	0.38	2.45	9132	2.01
GADF	Grid Tiling	All EEGs + EOG + ECG	1.05	0.24	2.10	7989	1.75
GASF	Channel Stacking	All EEGs + ECG	0.80	0.12	2.90	6263	1.34
GASF	Grid Tiling	All EEGs + EMG + ECG	0.55	0.03	3.90	4465	1.11
MTF	Channel Stacking	All EEGs + EOG	0.40	0.07	3.50	3992	1.01

To evaluate whether the leading strategies were significantly superior, a Mann–Whitney U test was conducted comparing the composite scores of the top five strategies against the remaining 1,011. Results indicated a highly significant difference ($p < 0.001$) with a rank-biserial correlation of 0.68, corresponding to a large effect size. This demonstrates that the top five methods substantially outperform the majority of strategies in terms of class separability and cluster compactness. Pairwise comparisons among the top five were performed using the Wilcoxon signed-rank test with Bonferroni correction ($\alpha = 0.005$). Significant differences were observed between GASF Grid Tiling with all channels and other methods. Other comparisons did not remain significant after correction, suggesting that while GASF Grid Tiling is consistently superior, the performance gap among the remaining top strategies is smaller.

To validate the practical utility of the top-ranked strategy, multiple ViT architectures were evaluated under five-fold stratified cross-validation. Among the tested models, ViT-B/32 achieved the highest performance with 83.1% accuracy and 79.4% macro-F1, closely followed by Swin-B and BEiT-B. Although the dataset comprised 108 recordings, this corresponds to over 100,000 30-second epochs, ensuring a statistically robust evaluation across diverse sleep stages. The achieved 83.1% accuracy is consistent with, and in several cases exceeds, results reported in comparable deep learning studies using similar datasets, indicating that this performance level is both realistic and clinically acceptable. The comparatively lower performance of DeiT-B and PVT-Small highlights differences in architectural design and feature extraction capabilities. Overall, these results demonstrate that the study successfully achieved its objective of identifying effective signal-to-image strategies for ViT-based sleep staging, with GASF grid tiling across all channels emerging as the most robust transformation. Beyond methodological insights, these findings have practical implications: by enabling more accurate and scalable automated sleep staging, the approach can reduce reliance on labor-intensive manual scoring, improve the detection of sleep disorders in clinical practice, and support large-scale research into sleep and brain health.

Summary and Conclusion

This study systematically evaluated over a thousand S2I combinations for ViT-based sleep stage classification, offering a novel perspective on representation separability. The GASF with grid tiling yielded the most discriminative embeddings and delivered 83.1% accuracy using ViT-B/32, outperforming typical CNN-based approaches reported in literature ($\approx 78\text{--}80\%$). These findings highlight the originality of our evaluation framework, which jointly considers statistical separability and classification benchmarks. The results demonstrate that optimal S2I selection can substantially enhance ViT performance for PSG-based sleep staging. By reducing reliance on manual scoring and improving scalability, the proposed strategy advances both methodological and clinical frontiers in automated sleep analysis.

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